

METRO TRANSIT SYSTEM

SQL Data Insights Report

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1. Network Overview

STATIONS 25 network-wide	TRAINS 20 across 4 route lines	REGISTERED PASSENGERS 465 15 without email on file	TOTAL BOOKINGS 1,490 448 unique riders
TOTAL FARE REVENUE Rs 146,092 all bookings, all-time	AVG FARE / BOOKING Rs 98.05 per ticket booked	AVG TRAVEL TIME 52.6 min per journey	REPEAT PASSENGERS 391 87.3% of active riders

2. Booking by Seat Class

Seat Class	Bookings	Revenue (Rs)	% of Revenue
General	1,200	Rs 117,362	80.3%
Premium	290	Rs 28,730	19.7%

General class accounts for the large majority of bookings (1,200 of 1,490), but Premium seating — at roughly a fifth of booking volume — still contributes 19.7% of total fare revenue, reflecting its higher per-ticket price point.

3. Revenue & Bookings by Route Line

Route Line	Bookings	Revenue (Rs)
Line 3	526	Rs 52,723
Line 5	439	Rs 42,024
Line 2	294	Rs 28,627
Line 4	231	Rs 22,718

Line 3 is the network's busiest corridor, generating Rs 52,723 from 526 bookings — the leading route line on both dimensions.

4. Top Performing Trains

Train Name	Bookings	Revenue (Rs)
Teal Line Express	84	Rs 8,390
Purple Line Local	84	Rs 8,529
Violet Line Local	81	Rs 7,883
Silver Line Express	81	Rs 8,739
Pink Line Local	79	Rs 7,869

Booking volume across all 20 trains is fairly even, ranging from 59 to 84 bookings per train — a spread of only 25 bookings. Teal Line Express and Purple Line Local share the top spot with 84 bookings each. **Teal Line Local** has the lowest activity in the network at 59 bookings, but this is a modest gap rather than a sign of an underused service.

5. Top Boarding Stations

Station	Bookings	Revenue (Rs)
Old Fort	70	Rs 7,235
Harbor Point	65	Rs 7,024
Metro Mall	65	Rs 6,716
Tech Park	72	Rs 6,681
Lakeview	64	Rs 6,559

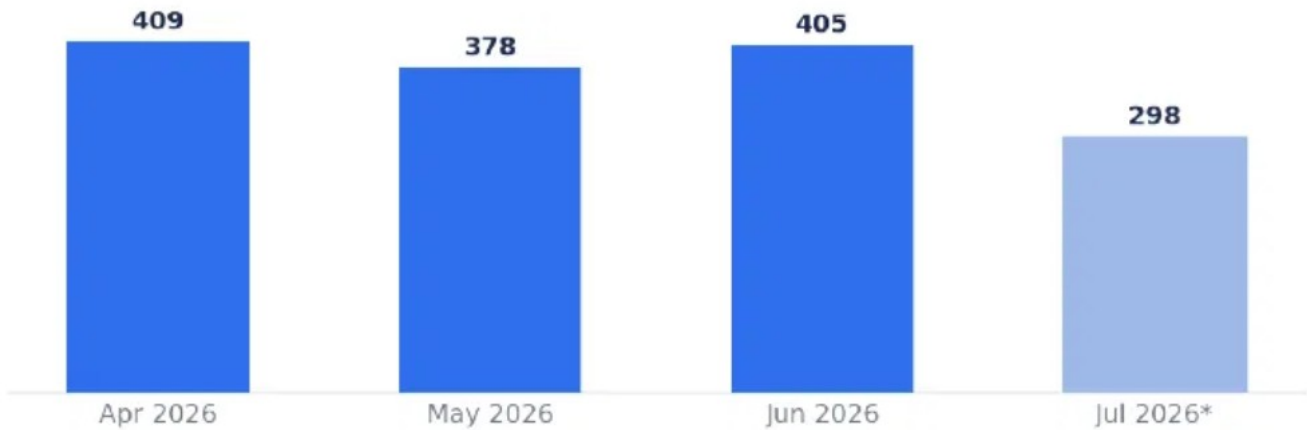
Old Fort (Chennai) is the strongest boarding point in the network, generating Rs 7,235 from 70 bookings. All 25 stations recorded at least one boarding and one destination trip, indicating the full station network is active with no unused nodes.

6. Fare Categories

Category	Definition	Bookings	Share
High Fare	fare > Rs 100	717	48.1%
Medium Fare	Rs 50 - Rs 100	459	30.8%
Low Fare	fare < Rs 50	314	21.1%

Nearly half of all bookings (48.1%) fall into the High Fare tier, indicating longer or premium journeys make up a substantial share of network demand rather than being a small niche.

7. Monthly Booking Trend



Bookings held steady between 2026-04-01 and 2026-07-20, averaging around 400 bookings per month from April through June 2026.
* July 2026 shows fewer bookings (298) because the dataset captures only a partial month (through July 20) rather than a genuine demand decline.

8. Passenger Demographics

MINIMUM AGE 8 years	MAXIMUM AGE 75 years	AVERAGE AGE 41.2 years, all passengers	PASSENGERS W/O EMAIL 15 of 465 registered
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Top Passenger Cities

City	Registered Passengers
Mumbai	69
Chennai	66
Hyderabad	62
Coimbatore	58
Delhi	56

9. Top 5 Passengers by Total Spend

Passenger	Trips	Total Fare Paid (Rs)
Ritu Patel	9	Rs 1,081.06
Anjali Verma	8	Rs 986.60
Suresh Gupta	9	Rs 948.35
Deepak Iyer	8	Rs 815.84
Naveen Sharma	8	Rs 791.36

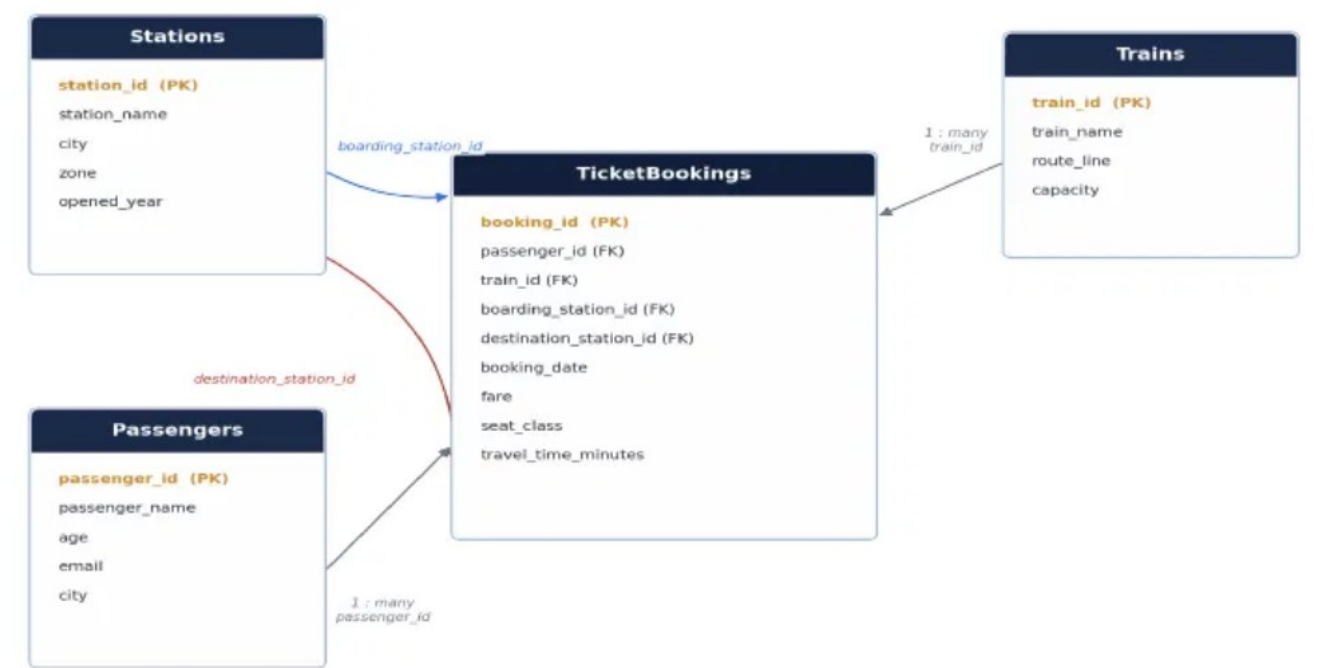
Ritu Patel is the network's highest-value passenger, with 9 trips totalling Rs 1,081.06 in fares — 2.7x the average rider's trip count.

10. Infrastructure Notes

- The network spans **25 stations** across 5 cities and 5 zones; every station has recorded both boarding and destination activity — no station in the network sits unused.
- **North zone** has the most stations (8), while **Central zone** has the fewest (3).
- The oldest stations in the network are **Lakeview** (Pune) and **University Gate** (Kochi), both opened in 1998.
- The newest station is **South Terminus** (Chennai), opened in 2022.
- Train capacity ranges from **180 to 320** seats, averaging **245.0** seats per train across the 20-train fleet.
- **Kochi** has the most stations of any single city (6), reflecting its role as a network hub.

11. Database Entity Relationship Diagram

TicketBookings is the central fact table of the schema, linking to Passengers and Trains, and referencing Stations **twice** — once for the boarding station and once for the destination station.



Gold fields – Primary Key | TicketBookings references Stations twice — once for boarding, once for destination

12. SQL Techniques Used

Database Design & Constraints

PRIMARY KEY • FOREIGN KEY • UNIQUE • DEFAULT • CHECK

Aggregation

COUNT() • SUM() • AVG() • COUNT(DISTINCT) • GROUP BY • HAVING

Text Functions

UPPER() • LENGTH()

Date & Time Functions

MONTH() • YEAR() • DAYNAME() • DATEDIFF() • DATE_ADD()

JOINS

INNER JOIN • LEFT JOIN • Multiple-table JOINS

Advanced SQL

Subqueries • CTEs • Views • CASE • Window Functions • RANK() • DENSE_RANK() • LAG() • LEAD()

Database Management

Transactions • ROLLBACK • Indexes • Composite Indexes

13. Key Business Insights

Route Performance:

Line 3 leads the network with Rs 52,723 in revenue from 526 bookings, while Line 4 trails all lines at Rs 22,718. Revenue tracks booking volume closely across all four lines.

Train Utilization:

Demand is well balanced across the 20-train fleet — booking counts range narrowly from 59 to 84, with no train showing signs of significant under-utilization.

Station Performance:

Old Fort in Chennai is the top revenue-generating boarding station (Rs 7,235), and all 25 stations show active boarding and destination traffic.

Fare Behavior:

High Fare bookings (over Rs 100) make up 48.1% of all bookings, more than Medium and Low Fare combined — the network's revenue base leans on higher-value journeys.

Premium Seating:

Premium seats are booked far less often than General (290 vs 1,200 bookings) but still deliver 19.7% of total revenue, confirming Premium fares carry a meaningful per-ticket price premium.

Passenger Behavior:

391 of 448 active riders (87.3%) have booked more than once, showing a strong base of repeat ridership rather than one-off usage.

Monthly Booking Pattern:

Booking volume was stable across April, May, and June 2026 (roughly 380-410 bookings/month); the lower July figure reflects a partial month of data rather than a decline in demand.

Revenue Concentration:

The top 5 passengers by spend contributed Rs 4,623.21 combined — a small group of frequent riders account for a disproportionate share of fare revenue.

14. Business Recommendations

Route & Capacity Planning:

Line 3 carries the heaviest booking and revenue load; prioritize it for peak-hour capacity reviews and maintenance scheduling ahead of lower-demand lines.

Premium-Seat Strategy:

Premium seating generates 19.7% of revenue from just 19.5% of bookings. Expanding Premium availability or targeted upsell promotions on high-traffic routes could lift per-booking revenue.

Station Resource Allocation:

Concentrate staffing and amenities at top-revenue boarding stations such as Old Fort, while continuing to monitor lower-traffic stations to ensure service levels stay adequate network-wide.

Passenger Retention:

With 87.3% of riders already returning, a loyalty or frequent-rider rewards program could convert more first-time passengers into repeat riders.

Fare & Revenue Optimization:

Since High Fare journeys already anchor 48.1% of bookings, consider bundled or discounted return-fare options to grow Medium and Low Fare segments without discounting the core High Fare base.

Data Quality:

15 passengers (3.2%) have no email on file, limiting the reach of retention campaigns; prioritize email capture at registration and ticket counters.

15. Executive Summary

- The Metro Transit network spans **25 stations** and **20 trains** across 4 route lines, serving **465 registered passengers**.
- The database recorded **1,490 bookings** generating **Rs 146,092** in total fare revenue, at an average of **Rs 98.05** per booking.
- **Line 3** is the strongest-performing route line, leading in both bookings and revenue.
- **Teal Line Express** and Purple Line Local share the top spot among trains, each with 84 bookings.
- **Old Fort** (Chennai) is the top-performing boarding station by revenue.
- Fare distribution skews toward higher-value trips: **48.1%** of bookings are High Fare (over Rs 100).
- **87.3%** of active riders are repeat passengers, indicating healthy ridership loyalty.
- Booking activity was stable across April-June 2026, with July's lower count reflecting a partial month of data.
- All 25 stations recorded genuine boarding and destination activity — the full network is in active use, with train capacity ranging 180-320 seats.

Overall Conclusion

This SQL analysis of the Metro Transit System demonstrates a well-utilized, financially healthy network: revenue is concentrated in a clear leading route line and station without any part of the network sitting idle, ridership shows meaningful repeat behavior, and fare revenue is anchored by higher-value journeys rather than a handful of outliers. The underlying relational structure — a single fact table (TicketBookings) connecting Passengers, Trains, and a dual reference to Stations — proved capable of supporting every operational question raised in this report, from route-level performance to individual passenger spend. With this foundation in place, the database can directly support ongoing decisions in capacity planning, station resourcing, fare strategy, and passenger retention.

16. SQL Query Appendix

A selection of queries demonstrating the core SQL techniques used to build this report.

Q16. Business Question	Find trains that have been booked more than 60 times.
SQL Concept	GROUP BY + HAVING
SQL Query	<pre>SELECT train_name, COUNT(DISTINCT booking_id) AS Total_bookings FROM Trains t INNER JOIN TicketBookings tb ON t.train_id = tb.train_id GROUP BY train_name HAVING Total_bookings > 60 ORDER BY Total_bookings DESC;</pre>
Result / Finding	All 20 trains in the fleet exceed 60 bookings — booking volume ranges from 59 to 84, confirming even utilization network-wide.

Q21. Business Question	Using a CTE, find the total fare collected per passenger.
SQL Concept	CTE
SQL Query	<pre>WITH TotalFare AS (SELECT passenger_id, SUM(fare) AS Total_fare FROM TicketBookings GROUP BY passenger_id) SELECT * FROM TotalFare ORDER BY Total_fare DESC;</pre>
Result / Finding	Ritu Patel tops the list with Rs 1,081.06 in total fare across 9 trips.

Q23. Business Question	Create a view combining booking, passenger, and train details.
SQL Concept	VIEW + Multi-table JOIN
SQL Query	<pre>CREATE VIEW DetailsPassenger AS SELECT tb.booking_id, p.passenger_name, t.train_name, tb.fare FROM Passengers p INNER JOIN TicketBookings tb ON p.passenger_id = tb.passenger_id INNER JOIN Trains t ON tb.train_id = t.train_id;</pre>
Result / Finding	Produces a single queryable view for booking-level reporting without repeating the underlying joins.

Q27. Business Question	Label each booking as High / Medium / Low Fare.
SQL Concept	CASE Statement
SQL Query	<pre>SELECT booking_id, fare, CASE WHEN fare > 100 THEN 'High Fare' WHEN fare BETWEEN 50 AND 100 THEN 'Medium Fare' ELSE 'Low Fare' END AS Fare_category FROM TicketBookings;</pre>
Result / Finding	High Fare bookings make up 48.1% of all 1,490 bookings — the single largest fare tier.

Q25. Business Question	Rank bookings by fare separately within each train.
SQL Concept	Window Function - DENSE_RANK() + PARTITION BY
SQL Query	<pre>SELECT booking_id, train_id, fare, DENSE_RANK() OVER (PARTITION BY train_id ORDER BY fare DESC) AS Rank_in_train FROM TicketBookings;</pre>
Result / Finding	Surfaces each train's highest-fare bookings independently, useful for identifying premium demand per train.

Q34. Business Question	Create a composite index on boarding and destination station.
SQL Concept	Composite Index
SQL Query	<pre>CREATE INDEX idx_boarding_destination ON TicketBookings (boarding_station_id, destination_station_id); SHOW INDEX FROM TicketBookings;</pre>
Result / Finding	Speeds up route-pair lookups (e.g. all bookings from a specific boarding station to a specific destination).